Student ID _____ Student Name _____

IS221: Web Applications Development

Faculty of Science, Technology and Environment
School of Computing, Information and Mathematical Sciences

Final Examination
Semester 1 2016

Face-To-Face and Online Mode

Duration of Exam: 3 hours + 10 minutes

Reading Time: 10 minutes

Writing Time: 180 minutes

INSTRUCTIONS

1. This examination has four sections:
   a. Section A: Multiple Choice (20 marks)
   b. Section B: Matching (20 marks)
   c. Section C: Coding (30 marks)
   d. Section D: Short and Long Answers (30 marks)
2. **All** questions are compulsory
3. Write **ALL** answers in the spaces provided within this paper.
4. This exam is **marked out of 100** and is worth **50%** of your **overall course marks**.
5. To **pass** this paper you require **at least 40% for this exam** which is equivalent to **40 marks out of 100**
6. This is a closed book exam.
7. This paper has 26 pages including this page.

Section A: Multiple Choice 20 Marks

Instructions:
Identify and circle the choice that best completes the statement or answers the question in the provided answer sheet on page 7. There is only **one** possible answer.

1) Two considerations if using graphics for site navigation are: keep them _____ and _____.

    a. simple, reuse them consistently
    b. simple, reuse them sparingly
    c. extravagant, don't reuse
    d. extravagant, complicated

2) There's a fine balance between a website being informative and it being overloaded with information. To limit information overload, a web designer should consider:

    a. Lengthy information segments, hypertext and control page length.
    b. Manageable information segments, lengthy pages and hypertext.
    c. Manageable information segments, control page length and hypertext.
    d. Manageable information segments, pages that requires multiple scrolls to reach the end of it and hypertext.

3) When designing navigation for mobile devices, three important considerations are:

    a. Minimize the number of clicks, regional navigation symbols and provide as many choices for the user as possible.
    b. Minimize the number of clicks, regional navigation symbols and simplify the user's choices.
    c. Minimize the number of clicks, universal navigation symbols and provide as many choices as possible for the user.
    d. Minimize the number of clicks, universal navigation symbols and simplify the user's choices.

Figure 1: Navigation bar

4) The navigation bar depicted in Figure1 is created using lists for navigation. To remove the default bullets and the default spacing that lists have the following style should be used:

   a. ul#navlist {padding-left: 0; margin-left: 0; list-style-type: none;}
   b. ul#navlist {padding-right: 0; margin-right: 0; list-style-type: none;}
   c. ul#navlist {padding-top: 0;margin-top: 0; list-style-type: none;}
   d. ul#navlist {padding-bottom: 0;margin-bottom: 0;list-style-type: none;}

5) Get and Post are two methods used to send form information back to the server for processing. If using the _____ method, information is send by including it in the URL which is _____ secure.

   a. Get, very
   b. Post, not
   c. Get, not
   d. Post, very

6) On Moodle, when submitting your assignments, you were able to select a file on your computer and uploaded it. This could be facilitated using _____.

   a. <output type="file" size="30“>
   b. <input type="file" size="30“>
   c. <upload type="file" size="30“>
   d. <download type="file" size="30“>

7) ASP.Net web forms are web pages identified with the file extension _____

   a. .css
   b. .aspx.vb
   c. .aspx
   d. None of the above

8) In three-tier client/server architecture, the _____ tier handles the interaction between the web browser and database server.

   a. Client
   b. Processing
   c. Data storage
   d. Authentication

9) <asp:TypeOfControl ID="ControlName" **Runat="Server"** /> adds a new control to the web form. The Runat="Server" attribute indicates that the control resides on the _____.

   a. server
   b. browser
   c. operating system
   d. network

10) <asp:Label ID="Label1" runat="server" ForeColor="Red"></asp:Label> will be rendered as:

   a.
   b. <label id="Label1" style="color:Red;"></label>
   c.
   d. <label id="Label11" style="color:Red;">

11) _____ are used by USP's website to share the same content among the multiple Faculty pages and facilitates ease of site maintenance, extension and modification. Respective faculties use _____ that enable them to still share common content but customize it with the faculty colours.

   a. Master Pages, Nested Master Pages
   b. Nested Master Pages, Master Pages
   c. Site Master, Nested Site Master
   d. Nested Site Master, Site Master

12) In the style declaration, p {color: red} – color:red is referred to as the _____ part of the Style Rule

    a. Selector
    b. Declaration
    c. Attribute
    d. Attribute value

13) A CSS rule that will assign font to be serif and centre aligned to h1 would be:

    a. h1{ text-align : centre ; font-family : serif;}
    b. h1{ text-align : centre ; font : serif;}
    c. h1{ text-align = " centre" ; font-family = "serif";}
    d. h1{ text-align = "centre" ; font = " serif";}

14) To set the font to be seriff for paragraphs, h1 and h2, the following style should be used:

    a. p, h1, h2 {font-family = "seriff";}
    b. p, h1, h2 {font-family : seriff;}
    c. p, h1, h2 {font-family=seriff;}
    d. None of the above

15) Given the style declarations below and the tag <h1> Welcome </h1>, Welcome will be colored _____

    External Stylesheet – h1{color: black;}
    Internal Stylesheet - h1 {color: red;}
    Inline style - h1 {color: green;}

    a. green
    b. red
    c. black
    d. colorless

16) a:link {color: #000000;} will result in:

    a. New links are black
    b. Active links are black
    c. Visited links are black
    d. All of the above

17) In CSS, ____ is an absolute measurement unit while _____ is a relative measurement unit.

    a. Em, pixel.
    b. Pixel, point
    c. Pixel, em
    d. Em, point

18) To get an image to be displayed in the top right hand corner of the page, the _____ property of the image should be set to _____

    a. Flow, right
    b. Float, right
    c. Clear, right
    d. None of the above

19) In responsive web pages, _____ are used to determine the stylesheet that will best render the website for the remote device being used to access the website.

    a. CSS script
    b. CSS template
    c. CSS queries
    d. CSS media queries

20) When using a Master Page for the layout of a web page, the _____ controls identify locations where content may be inserted in a content page.

    a. ContentPlaceHolder
    b. FormContentHolder
    c. PagePlaceHolder
    d. DivisionPlaceHolder

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Section A: Answer sheet

Circle the correct answer.

1.	**a**	**b**	**c**	**d**
2.	**a**	**b**	**c**	**d**
3.	**a**	**b**	**c**	**d**
4.	**a**	**b**	**c**	**d**
5.	**a**	**b**	**c**	**d**
6.	**a**	**b**	**c**	**d**
7.	**a**	**b**	**c**	**d**
8.	**a**	**b**	**c**	**d**
9.	**a**	**b**	**c**	**d**
10.	**a**	**b**	**c**	**d**
11.	**a**	**b**	**c**	**d**
12.	**a**	**b**	**c**	**d**
13.	**a**	**b**	**c**	**d**
14.	**a**	**b**	**c**	**d**
15.	**a**	**b**	**c**	**d**
16.	**a**	**b**	**c**	**d**
17.	**a**	**b**	**c**	**d**
18.	**a**	**b**	**c**	**d**
19.	**a**	**b**	**c**	**d**
20.	**a**	**b**	**c**	**d**

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Section B: Matching Questions 20 Marks

Write the letter of the correct term that matches the given description in the space provided

A. *Contextual link*

B. *<asp:TextBox id="txtAddr" TextMode="MultiLine" Runat="server" />*

C. *Text-based linking*

D. *Server side scripting*

E. *HTML List elements*

F. *ul#navlist li{ display: inline; }*

G. *_EVENTTARGET and _EVENTARGUMENT*

H. *Box model*

I. *RepeatColumns*

J. *Overflow*

K. *Clear*

L. *Padding*

1. _____ Most effective way to provide navigation on site
2. _____ Allow users to jump to related ideas by clicking the word of interest
3. _____ Creates a horizontal navigation bar
4. _____ Used for User feedback text box
5. _____Hidden fields used by the _doPostBack function to assign values passed from the hyperlink
6. _____ Preferred element for containing navigation links
7. _____ Logging into Facebook and Moodle
8. _____ Space between actual content and border
9. _____ Content is larger than the area that contains it
10. _____ Rectangular boxes containing content on a web page
11. _____Number of columns of Checkboxes to display
12. _____ Used to force the caption for a picture that was floated left to appear beneath the picture.

M. *Float*

N. *img {opacity: .75;}*

O. *Column drop*

P. *PNG*

Q. *Broken layout*

R. *@media screen and (max-width: 480px) { header {width: 90%;} }*

S. *Breakpoint*

T. *div.wrapper { width: 100%; min-width: 750px; max-width: 1220px; }*

13. _____ Floating a picture which extends beyond its containing element.

14. _____ Control compression and expansion of content in a flexible layout

15. _____ 758-pixel threshold

16. _____ Applying conditional style based on outcome of media query

17. _____ Sets the transparency of an image

18. _____ total width of columnar element exceeds the width of containing element.

19. _____ Supports Transparency and interlacing but not animation

20. _____ Forcing an image to appear at the top right hand corner of a page

Section C: Coding Questions 30 Marks

You have been employed to revamp USP's Sports and Recreation website. Currently gym membership subscription is a paper based process which needs to be made available online for potential members to fill in the web based forms.

Question 1

Given the code for the student gym member registration page depicted in Listing 1 on pages 11, 12 and 13, draw the web form when it is rendered.

(10 marks)

Listing 1 – Web page code

```
<%@ Page Title="Home Page" Language="VB" MasterPageFile="~/Site.Master"
AutoEventWireup="false"
    CodeFile="Subscribe.aspx.vb" Inherits="_Default" %>

<asp:Content ID="HeaderContent" runat="server" ContentPlaceHolderID="HeadContent">
    <style type="text/css">
        .style1
        {
            color: #0033CC;
            text-align: center;
            font-family: FangSong;
        }
        .style2
        {
            width: 600px;
        }
        .style3
        {
            height: 38px;
        }
        .style5
    {
        font-family: FangSong;
        color: #0033CC;
    }
    </style>
</asp:Content>
<asp:Content ID="BodyContent" runat="server" ContentPlaceHolderID="MainContent">
    <h2 class="style1">
        Student Gym Registration</h2>
    <p class="style1">
        <asp:Image ID="Image2" runat="server"
            ImageUrl="~/Images/UniversityFitnessCentre.png" Height="200px"
            Width="100%" />
    </p>
    <p class="style1">
         </p>
    <table class="style2" align="center">
        <tr>
            <td>
            Gender:</td>
            <td>
                <asp:RadioButtonList ID="rblGender" runat="server" RepeatColumns="2"
                    TextAlign="Left">
                    <asp:ListItem>Female</asp:ListItem>
                    <asp:ListItem>Male</asp:ListItem>
                </asp:RadioButtonList>
            </td>

        </tr>
        <tr>
         <td>
                First Name:</td>
```

```
            <td>
                <asp:TextBox ID="txtFirstName" runat="server"
Width="160px"></asp:TextBox>
            </td>
        </tr>
        <tr>
            <td>
                Family Name:</td>
            <td>
                <asp:TextBox ID="txtFamilyName" runat="server"
Width="160px"></asp:TextBox>
            </td>
        </tr>
        <tr>
            <td>
                Student ID:</td>
            <td>
                <asp:TextBox ID="txtStudID" runat="server" Width="160px"></asp:TextBox>
            </td>
        </tr>
        <tr>
            <td>Interests:
            </td>
            <td>
                <asp:TextBox ID="TextBox1" runat="server"
TextMode="MultiLine"></asp:TextBox>
            </td>
        </tr>
        <tr>
            <td>
                Date of Birth DD/MM/YY:</td>
            <td>
                <asp:TextBox ID="txtDOB" runat="server" Width="160px"
TextMode="Date"></asp:TextBox>
            </td>
        </tr>
        <tr>
            <td>Phone:</td>
            <td>
                <asp:TextBox ID="txtPhone" runat="server" TextMode="Phone"></asp:TextBox>
            </td>
        </tr>
        <tr>
            <td>Email:</td>
            <td>
                <asp:TextBox ID="txtEmail" runat="server" Width="150px"
TextMode="Email"></asp:TextBox>
            </td>
        </tr>
        <tr>
            <td>Emergency Contact Name:
            </td>
            <td>
                <asp:TextBox ID="txtECName" runat="server"></asp:TextBox>
            </td>
        </tr>
```

```
        <tr>
            <td>Emergency Contact Phone Number:
            </td>
            <td>
                <asp:TextBox ID="txtECNumber" runat="server"
TextMode="Phone"></asp:TextBox>
            </td>
        </tr>
        <tr>
            <td>
                <h2 class="style5">Subscription Period:</h2></td>
            <td>
                <asp:DropDownList ID="ddlSubscriptionTerm" runat="server">
                    <asp:ListItem>Semester 1</asp:ListItem>
                    <asp:ListItem>Semester 2</asp:ListItem>
                    <asp:ListItem>Whole Year</asp:ListItem>
                </asp:DropDownList>
                <asp:XmlDataSource ID="xdsSubscriptions" runat="server"
                    DataFile="~/App_Data/Subscriptions.xml"></asp:XmlDataSource>
            </td>
        </tr>
        <tr>
            <td>
                <h2 class="style5">Subscription For:</h2></td>
            <td>
                <asp:CheckBoxList ID="cblSubscriptions" runat="server"
                 DataSourceID="xdsSubscriptions" DataTextField="name"
              DataValueField="name">
                </asp:CheckBoxList>
            </td>
        </tr>
        <tr>
            <td class="style3">
                </td>
            <td class="style3">
                <br />
                <asp:Button ID="btnSubscribe" runat="server" Text="Subscribe Me"
                    style="text-align: center" ForeColor="#0033CC" />
            </td>
        </tr>
    </table>
    <p>
        <asp:Label ID="lblVerify" runat="server" ForeColor="#0033CC"></asp:Label>
        <asp:AccessDataSource ID="adsSubscribers" runat="server"
              (Code omitted for SQL queries) </asp:AccessDataSource> </p>

</asp:Content>
```

Question 2

Analyse the use of CSS in the Style section on the code provided in Listing1 on pages 11, 12 and 13. Write a Style that will have all headings and paragraphs to be colored blue, centre aligned and font to be Calibri and assign background color to grey. **(5 marks)**

Question 3

The Event Handler code for the student gym subscription page should extract all the student's data from the form and insert them into the respective tables in the Subs database. **(15 marks)**

a) The details of the Subs database are depicted in Figure 2, however there are some issues with it. With reference to the code given in Listing 1 on pages 11, 12 and 13 for the web form and figure 2, determine the potential issues that the Subs database has and redesign the Subs database to address these issues and ensure that the students' gym subscription details are fully captured. **(5 marks)**

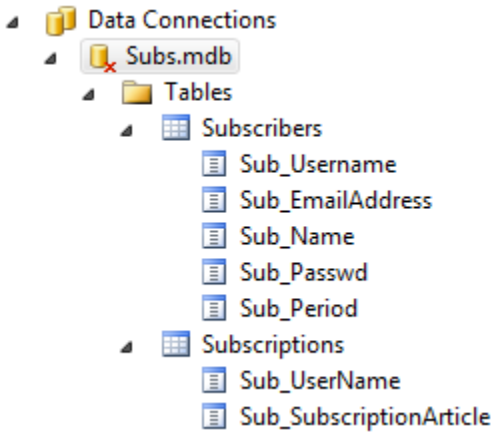

Figure 2: Subs database structure

Identified issue

_____
_____
_____
_____
_____
_____
_____
_____
_____
_____
_____

Database Tables design

b) Complete the Event Handler code below to successfully extract data from the Student Gym Registration form and insert them into the Subscriber and Subscription tables. **(10 marks)**

```
Partial Class _Default
 Inherits System.Web.UI.Page

Protected Sub btnRegister_Click(ByVal sender As Object, ByVal e As
System.EventArgs) Handles btnSubscribe.Click

'add a record in the Subscriptions table for each article subscribed per user
```

Section D: Short/Long Answer Questions 30 Marks

Question 1

Effective navigation should provide information that helps the user to answer the following questions:

- Where am I?
- Where can I go?
- How do I get there?
- How do I get back to where I started?

Compare and contrast how Digicel, Vodafone and INKK implemented their navigation schemes and how they provided answers to the stated questions. For each question determine the required information to be provided and how each of the sites provided the answers.

(10 marks)

<u>Where am I?</u>

Required information

_____

_____

_____

Digicel

_____

_____

_____

Vodafone

_____

_____

_____

INKK

_____

_____

_____

Where can I go?

Required information

_____
_____
_____

Digicel

_____
_____
_____

Vodafone

_____
_____
_____

INKK

_____
_____
_____

How do I get there?

Required information

_____
_____
_____

Digicel

_____
_____
_____

Vodafone

_____
_____
_____

INKK

_____
_____
_____

How do I get back to where I started?

Required information

_____
_____
_____

Digicel

_____
_____
_____

Vodafone

_____
_____
_____

INKK

_____
_____
_____

Question 2

Masterpages and CSS play fundamental roles in the implementation of websites. With reference to your assignment 2, outline how you used Masterpages and CSS to create your website.

(5 marks)

_____
_____
_____
_____
_____
_____
_____
_____
_____

Question 3

ASP.NET applications processing is depicted in figure3 . Using the labels 1 – 5, and your Student Gym registration form describe this process.

(5 marks)

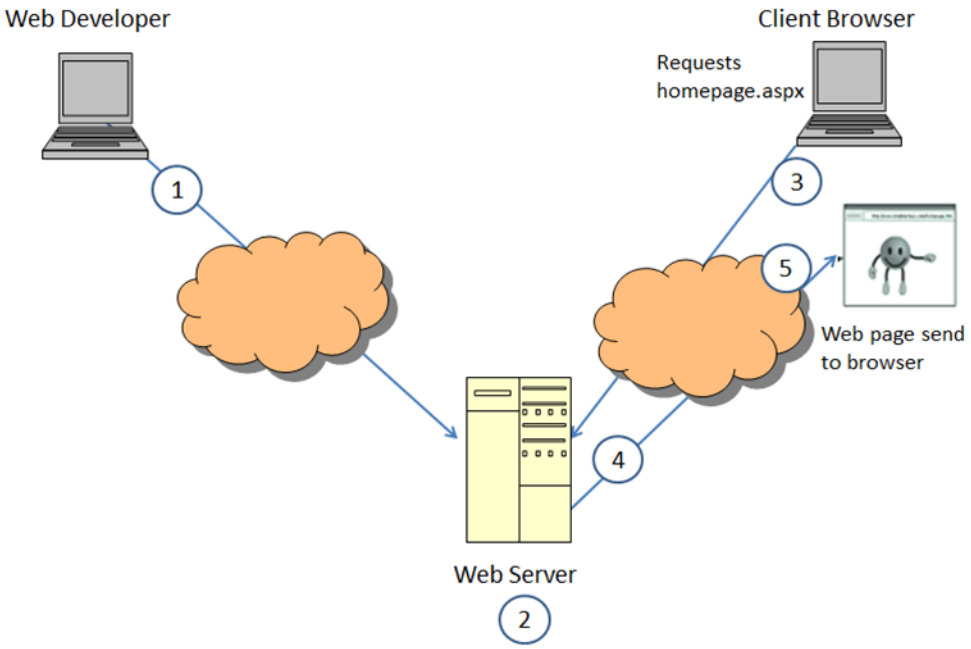

Figure 3: ASP.Net applications processing

_____
_____
_____
_____
_____
_____
_____
_____
_____
_____
_____

Question 4

Responsive web pages are pages that respond to how users are accessing it and displays accordingly. CSS media queries, flexible images and flexible layouts are elements that help facilitate responsive webpage designs. With reference to your assignment 2 and the stated elements that facilitate responsive webpage designs, how would you design a responsive version of your homepage for mobile devices that have screen sizes ranging from 320px to 768px? Justify your design choice. **(10 Marks)**

_____
_____
_____
_____
_____
_____
_____
_____
_____
_____

<u>Current Homepage</u>

Proposed Responsive Homepage design for mobile device

Student ID_____ Student Name_____

Web form Draft Page

THE END